



Scythe Yasya with its unique fin arrangement

sipate all the heat and again, we see a drop in cooling efficiency. With passive heatsinks, the fin spacing is extremely important since there are no cooling fans to channel air through the fins. The fin spacing is a factor of the material used for the fins, the length of the fins and the fin thickness. With active heat sinks, i.e. ones with fans, the factors are still the same but you can have narrower spacing.

One thing that used to be a common gimmick was the fin shape. There were coolers with outlandishly designed fins which claimed to aid efficiency. Thankfully, we've stopped seeing such designs and most of the heatsinks we now see have flat rectangular fins.

Coolers with multiple towers are not gimmicky though. Having multiple cooling towers is actually more efficient for simple reasons. Multiple towers are only used if there are a lot of heat pipes being used. With multiple towers, you are increasing the amount of fins attached to each heat pipe. This proportionally increases the surface area and thus the cooling efficacy of the CPU cooler.

Fans

The last major component of the CPU cooler is the fan. Cooling fans have a few important characteristics that you need to keep in mind. However, there aren't any CPU coolers out there which don't come with fans included. So it becomes a really expensive affair if you're getting a custom CPU cooler and third-party fans separately.

The longevity of the cooling fans are dependent on the bearing. If the bearing has been precisely machined and aligned properly, there will be no wobbling. Wobbling on the other hand is harmful because it causes intense vibration at high RPM and that can cause components to come apart in your PC. This is true with fans that have heavy blades. Just Google for broken fan blades and you'll come across videos that'll make this a lot clearer. An unbalanced fan is the same thing.